

MODEL TRAINING REPORT

Project: Multi-Class Skin Disease Classification & Diagnosis Platform

Dataset: HAM10000

Framework: TensorFlow / Keras

1. Training Overview

The objective of model training was to develop an accurate deep learning model capable of classifying dermoscopic skin lesion images into seven disease categories. Multiple convolutional neural network (CNN) models and transfer learning architectures were trained and evaluated to identify the best-performing model. The training process involved image preprocessing, data augmentation, class balancing using class weights, optimization using different learning strategies, and model evaluation using several performance metrics.

2. Models Trained

Model	Description
Basic CNN	Baseline convolutional neural network
CNN + Batch Normalization	Improved training stability
Deep CNN	Additional convolution layers
MobileNetV2	Transfer Learning
ResNet50	Transfer Learning
DenseNet121	Transfer Learning
EfficientNetB0	Transfer Learning + Fine Tuning

3. Training Configuration

The models were trained using TensorFlow and Keras. Images were resized to 128×128 pixels before being supplied to the neural networks. Categorical cross-entropy was used as the loss function because the problem involves multi-class classification. Different optimizers including Adam and RMSprop were evaluated during training. Early Stopping was employed to monitor validation loss and prevent unnecessary epochs, while learning rate scheduling was used to improve convergence. Class weights were applied to compensate for the imbalanced distribution of disease categories within the HAM10000 dataset.

4. Hyperparameter Settings

Parameter	Value
Image Size	128×128
Batch Size	32

Optimizer	Adam / RMSprop
Loss Function	Categorical Crossentropy
Epochs	50
Callbacks	EarlyStopping
Learning Rate	0.001
Class Weights	Enabled

5. Transfer Learning

Transfer learning was employed using pretrained convolutional neural networks initialized with ImageNet weights. Initially, the pretrained feature extractor layers were frozen while the custom classification head was trained. During fine-tuning, selected upper layers of the pretrained network were unfrozen, allowing the model to learn domain-specific skin lesion features while retaining useful visual representations learned from large-scale datasets. Among all transfer learning models, EfficientNetB0 achieved the best overall classification performance.

6. Final Model Performance

Metric	Value
Accuracy	72.39 %
Precision	68.35 %
Recall	72.39 %
F1 Score	68.21 %
ROC-AUC	83.49 %

7. Conclusion

The model training process successfully produced several deep learning models for skin disease classification. Extensive experimentation with different CNN architectures, transfer learning models, hyperparameter settings, and optimization strategies resulted in significant performance improvements. Fine-tuned EfficientNetB0 achieved the highest classification accuracy among all evaluated models. The trained model was subsequently integrated into the Flask web application to provide real-time skin disease prediction, confidence estimation, probability distribution, and Grad-CAM visual explanations.